SPAGHETTILAB | BACKBONE CARRIER

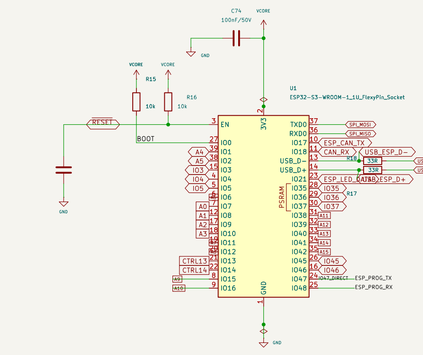
Physical IC groups | all units of each IC together | named nets preserve functional links

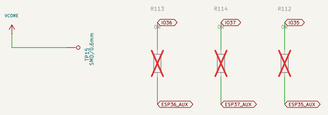
P1–P6 = external experiment rows | A0–A5 = ESP-side signals | J3 = separate direct access. Named arrows connect blocks.

READING GUIDE: Q3–Q8 = power stages | U12/U13 = signal isolation | J1/J140 = shared headers | R56–R61 = shunts

U1 | OPTIONAL ESP32-S3

Adds Wi-Fi, Bluetooth and familiar ESP32 development tools.

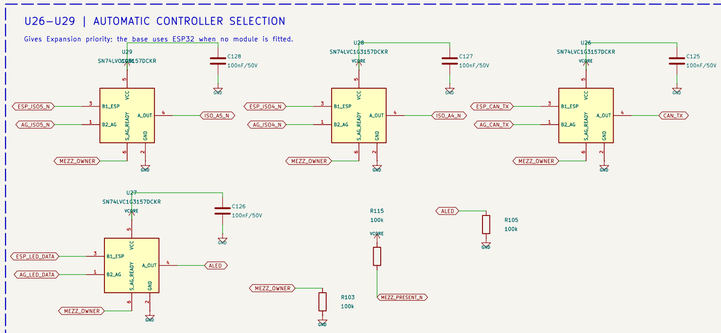




GPIO35/36/37 AUX links are DNP PSRAM guards. Fit only on compatible ESP32 variants. USB does not pass through FPGA logic.

U26–U29 | AUTOMATIC CONTROLLER SELECTION

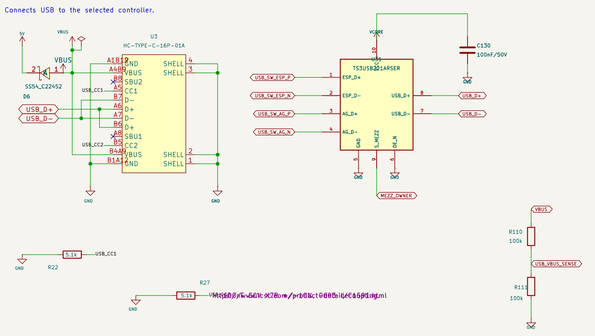
Gives Expansion priority: the base uses ESP32 when no module is fitted.



MEZZ_OWNER LOW = ESP32, HIGH = Expansion. R103 = 100k pull-down: Expansion adds R109 = 10k pull-up. Claim is not firmware-ready. USB, CAN TX, RGB and ISO4/5 are selected.

USB-C + U31 | COMPUTER CONNECTION

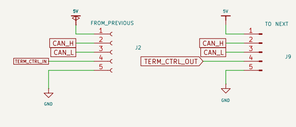
Connects USB to the selected controller.



U31: LOW selects ESP32, HIGH selects Expansion. Both USB series-resistor pairs are fitted. Owner changes require USB detach and re-enumeration.

J2 / J9 | MAGNETIC CAN LINK

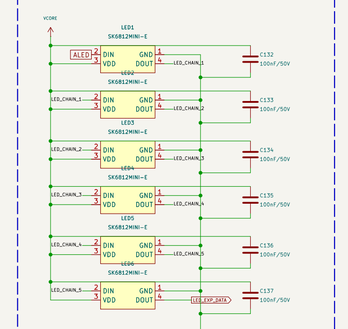
Passes power and messages between neighbouring backbones.



Q9/Q10 detect both neighbours by TERM current. Q2/U8 insert 120R at either end. No MCU.

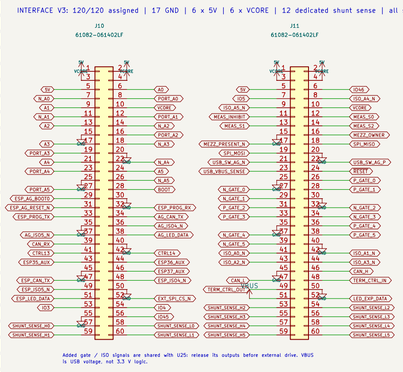
LED1–LED6 | PORT STATUS

Six configurable ports. Header and serial LED order: P1–P6.



J10 / J11 | EXPANSION INTERFACE – 2 x 60 CONTACTS

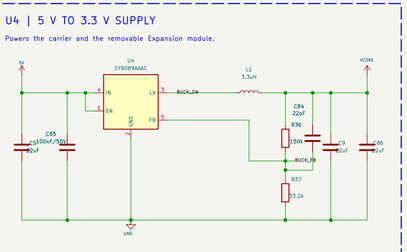
INTERFACE V3: 120/120 assigned | 17 GND | 6 x 5V | 6 x VCORE | 12 dedicated shunt sense | all services retained.



Added gate / ISO signals are shared with U25: release its outputs before external drive. VBUS is USB voltage, not 3.3 V logic.

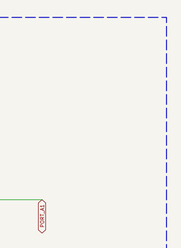
U4 | 5 V TO 3.3 V SUPPLY

Powers the carrier and the removable Expansion module.



U25 | PORT CONFIGURATION

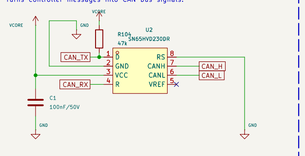
Shared I2C control of the six power stages.



U25 + C129: common to all variants. One active I2C master at a time. U12/U13 are grouped separately below.

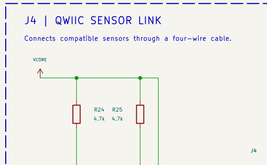
U2 | CAN TRANSCEIVER

Turns controller messages into CAN bus signals.



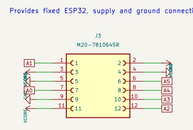
J4 | QWIIC SENSOR LINK

Connects compatible sensors through a four-wire cable.



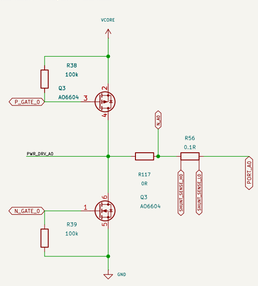
J3 | DIRECT ESP32 ACCESS

Provides fixed ESP32, supply and ground connections.



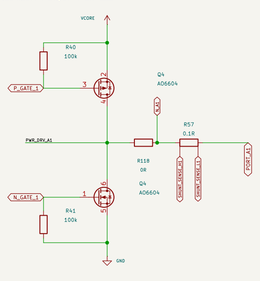
P1 | Q3 POWER STAGE

Dual MOSFET + R38/R39 + R56 shunt



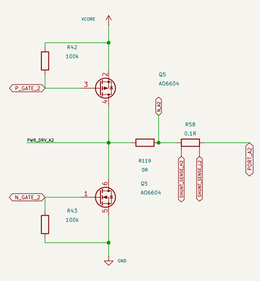
P2 | Q4 POWER STAGE

Dual MOSFET + R40/R41 + R57 shunt



P3 | Q5 POWER STAGE

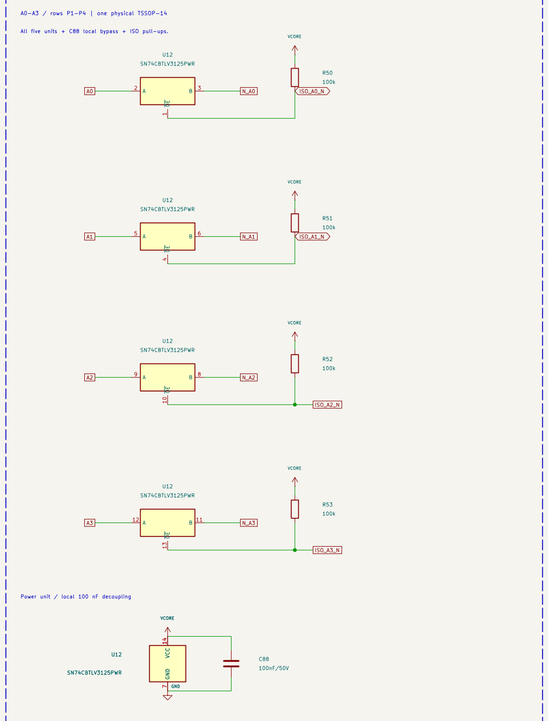
Dual MOSFET + R42/R43 + R58 shunt



U12 | SN74CBTLV3125 SIGNAL ISOLATION

A0–A3 / rows P1–P4 | one physical TSSOP-14

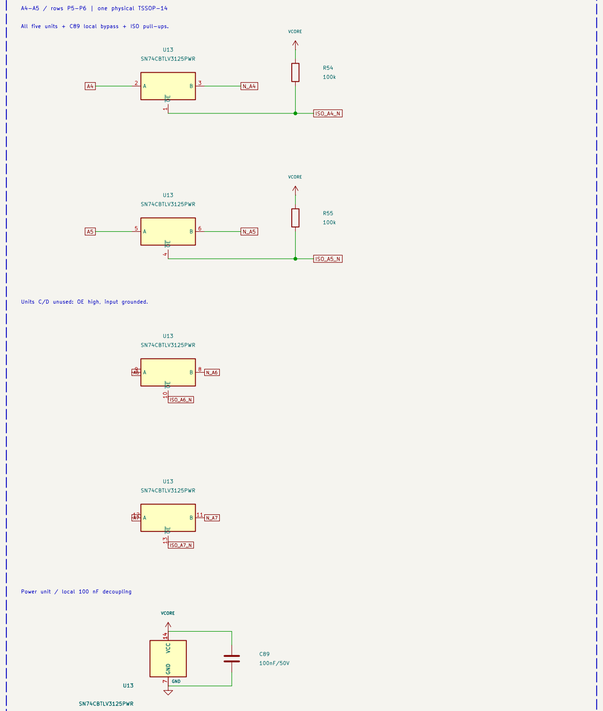
All five units + C88 local bypass + ISO pull-ups.



U13 | SN74CBTLV3125 SIGNAL ISOLATION

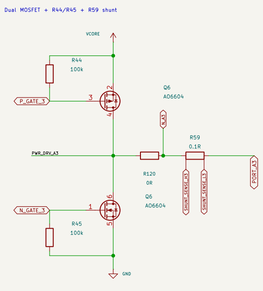
A4–A5 / rows P5–P6 | one physical TSSOP-14

All five units + C89 local bypass + ISO pull-ups.



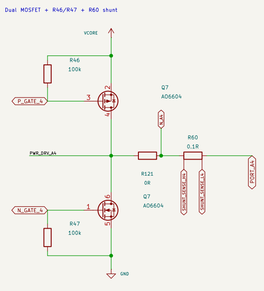
P4 | Q6 POWER STAGE

Dual MOSFET + R44/R45 + R59 shunt



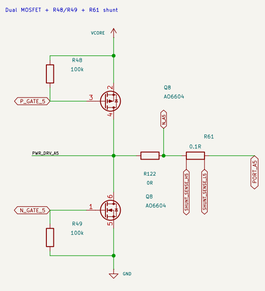
P5 | Q7 POWER STAGE

Dual MOSFET + R46/R47 + R60 shunt



P6 | Q8 POWER STAGE

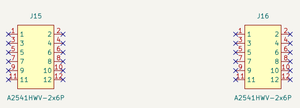
Dual MOSFET + R48/R49 + R61 shunt



Rows switch 3V3 / GND / Hi-Z. Never enable both MOSFETs together: set ESP GPIO Hi-Z before switching.

CJT CONNECTOR FIT TEST | J15 / J16

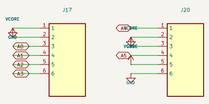
Unconnected mechanical samples. Not fitted in production.



2 x 6 contacts | 2.54 mm pitch | body 15.74 x 5.00 x 8.50 mm

BKL 2x8 DUAL ENTRY | SENSOR & CONFIGURABLE HEADERS

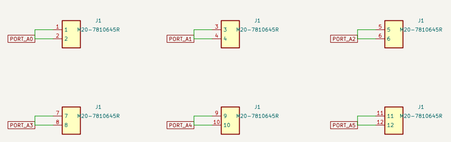
J17 / J20: direct ESP sensor rows. A0–A5 plus mixed 3V3 / GND. J18 / J19: configurable P1–P6.



Bottom insertion | 20.72 x 5 x 5 mm | land pattern requires physical qualification

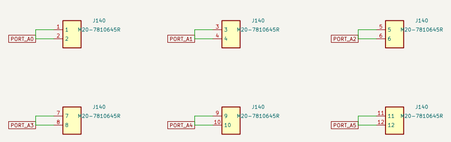
J1 | CONFIGURABLE ROW HEADER

All six units belong to this single connector.



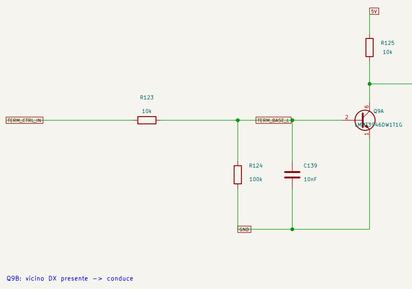
J140 | CONFIGURABLE ROW HEADER

All six units belong to this single connector.

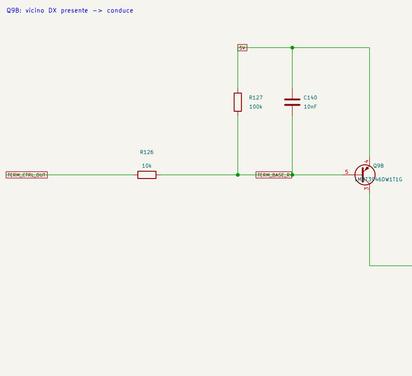


TERM AUTOMATICO A CORRENTE | Q9: rilevatori NPN+PNP | Q10: logica NPN+NPN | D7: due diodi OR

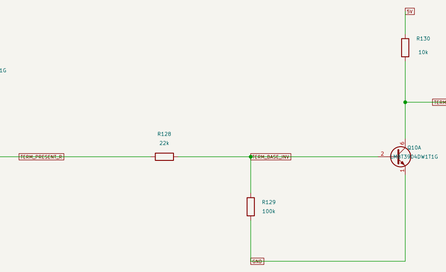
Q9A: vicino SX presente –> conduce

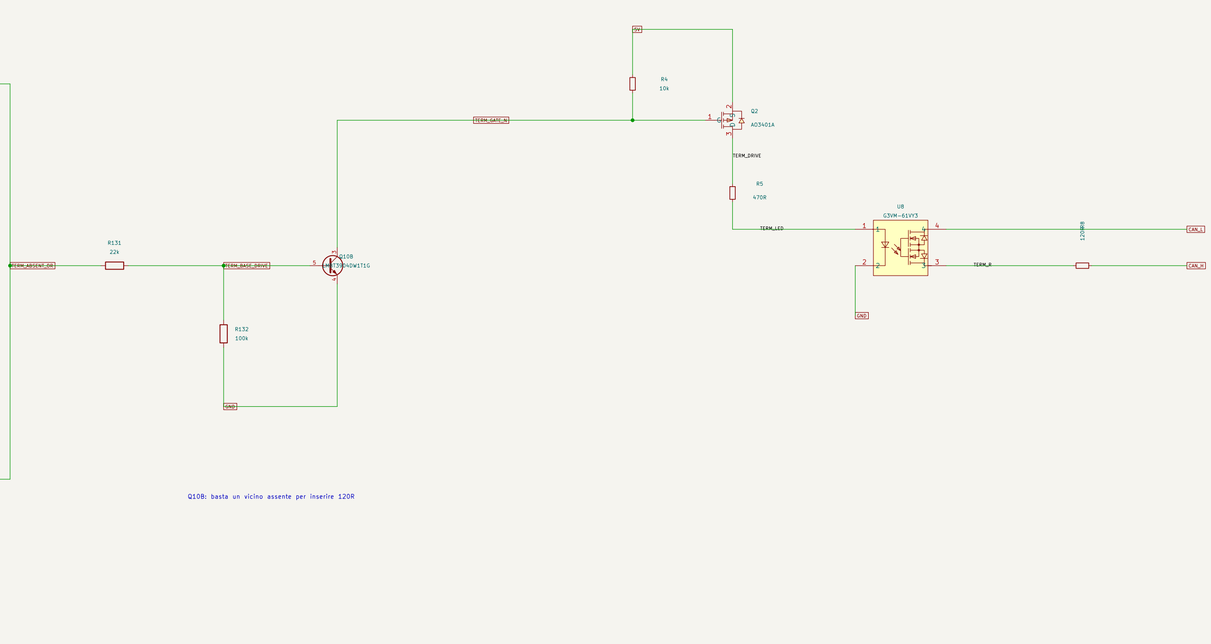


Q9B: vicino DX presente –> conduce



Q10A inverte il rilevamento destro





Q10B: basta un vicino assente per inserire 120R

Nessun micro necessario. TERM DX di una scheda si collega a TERM SX della successiva. Corrente di link circa 0.18mA a 5V.

C139/C140 filtrano le basi. R e C: 0603. Tutte le schede devono usare questa revisione: non pilotare TERM dalla mezzanina.

Simulazione con modelli generici: non qualifica hardware. Verificare contatti, avvio, ESD e CAN sul prototipo.

16-row extension and slow controls

File: ports16.kicad_sch

16-GPIO VARIANT: service controls moved to U32/U33/U34.
GPIO35–37 and boot straps are not used as experiment signals.
See ports16 sheet. Firmware update required; original design retained separately.

Functional clusters; wiring preserved. Not production-qualified.

Sheet: /
File: backbone.kicad_sch

Title: SpaghettiLAB – Backbone carrier

Size: User | Date:
KiCad E.D.A. 10.0.3
Rev: Mezzanine V1 – organ
Id: 1/2

U32: I2C 0x21 / port controls

U33: I2C 0x22 / port controls

U34: I2C 0x23 / slow services; presence input

PORT 7 / L7 / GPIO6

PORT 8 / L8 / GPIO11

PORT 9 / R1 / GPIO12

PORT 10 / R2 / GPIO15

PORT 11 / R3 / GPIO16

GPIO switch: U13C on main sheet

GPIO switch: U13D on main sheet

PORT 12 / R4 / GPIO38

PORT 13 / R5 / GPIO39

PORT 14 / R6 / GPIO40

PORT 15 / R7 / GPIO41

PORT 16 / R8 / GPIO42

LEFT L1–L8 / 3 contacts per independent node

RIGHT R1–R8 / 3 contacts per independent node

VCORE = 3.3 V. Disable GPIO path before selecting a rail. Never enable both MOSFETs.
No crosspoint matrix. Firmware and mezzanine interface revision required.
Existing six shunts retained: ten added sense pairs are local only until the interface is redesigned.

Sheet: /16-row extension and slow controls/
File: ports16.kicad_sch

Title:

Size: User | Date: | Rev:

KiCad E.D.A. 10.0.3 | Id: 2/2